SOP: Resolution of BGP Route Churn, OSPF Adjacency Loss, and ISIS LSP Flooding on Huawei Access Node (JAY-ACC-39)

Node: JAY-ACC-39 | **Vendor:** Huawei | **Severity:** Critical | **Alarm Code:** ALARM_8725

Description

This Standard Operating Procedure (SOP) outlines the steps to resolve the failure of BGP route churn, OSPF adjacency loss, and ISIS LSP flooding on Huawei Access Node JAY-ACC-39 at the Jayanagar Metro Site. The failure is likely caused by a misconfiguration or hardware issue related to the routing protocols on the Access Node. The Huawei Access Node is responsible for routing traffic between multiple networks and protocols, and the failure can lead to significant disruptions in network connectivity. The root cause of the failure is likely related to the BGP routing protocol, which is responsible for exchanging routing information between the Access Node and other routers in the network. The OSPF and ISIS protocols may also be affected due to their dependency on BGP for routing information. The failure can be caused by a number of factors, including misconfiguration of BGP peers, incorrect routing policy configuration, or hardware issues such as faulty network interface cards or memory errors. The resolution steps outlined in this SOP aim to identify and correct the root cause of the failure, ensuring that the network is restored to a stable and operational state.

Pre-requisite Checks

- Verify that the site location is the Jayanagar Metro Site and the Node ID is JAY-ACC-39
- Check the status of redundant links to determine if they are active and functional
- Ensure that all relevant network equipment is powered on and functioning properly
- Dispatch remote hands to the site location to assist with physical checks and repairs if necessary

Potential Root Causes

Cause	Probability
Misconfigured BGP peers	High
Incorrect routing policy configuration	Medium
Hardware issues such as faulty network interface cards or memory errors	Low

Diagnostic Commands & Expected Outputs

```
Command: display bgp peer, Expected Output: Display BGP peer information, Description: This command displays information about BGP peers, including the peer ID, IP address, and state. It helps to identify any misconfigured BGP peers or issues with BGP neighbor relationships.
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```
Command: display ospf neighbor, Expected Output: Display OSPF neighbor information, Description: This command displays information about OSPF neighbors, including the neighbor ID, IP address, and state. It helps to identify any issues with OSPF adjacency or neighbor relationships.

Command: display isis neighbor, Expected Output: Display ISIS neighbor information, Description: This command displays information about ISIS neighbors, including the neighbor ID, IP address, and state. It helps to identify any issues with ISIS adjacency or neighbor relationships.
```

Resolution Steps

Step 1: Step: Verify BGP peer configuration, Command: display bgp peer, Description: Verify the BGP peer configuration to ensure that all peers are correctly configured and functioning properly. Check for any misconfigured peers or issues with BGP neighbor relationships.

Step 2: Step: Verify OSPF and ISIS neighbor relationships, Command: display ospf neighbor; display isis neighbor, Description: Verify the OSPF and ISIS neighbor relationships to ensure that all neighbors are correctly configured and functioning properly. Check for any issues with OSPF or ISIS adjacency or neighbor relationships.

Step 3: Step: Correct misconfigured BGP peers, Command: undo bgp peer , Description: Correct any misconfigured BGP peers by removing the incorrect configuration and re-establishing the correct peer relationship.

Step 4: Step: Update routing policy configuration, Command: set routing-policy , Description: Update the routing policy configuration to ensure that it is correct and functioning properly. Check for any issues with routing policy configuration or implementation.

Step 5: Step: Replace faulty hardware, Command: replace , Description: Replace any faulty hardware components, such as network interface cards or memory, to ensure that the network is stable and functioning properly.

Post-Check Verification

```
Command: display bgp peer, Expected Output: Display BGP peer information, Description: Verify that the BGP peer configuration is correct and functioning properly. Check for any issues with BGP neighbor relationships or misconfigured peers.

Command: display ospf neighbor, Expected Output: Display OSPF neighbor information, Description: Verify that the OSPF neighbor relationships are correct and functioning properly. Check for any issues with OSPF adjacency or neighbor relationships.

Command: display isis neighbor, Expected Output: Display ISIS neighbor information, Description: Verify that the ISIS neighbor relationships are correct and functioning properly. Check for any issues with ISIS adjacency or neighbor relationships.
```

Roll-back Procedures

- Step: Restore original BGP peer configuration, Command: undo undo bgp peer , Description: Restore the original BGP peer configuration by re-establishing the incorrect configuration.

- Step: Restore original routing policy configuration, Command: undo set routing-policy , Description: Restore the original routing policy configuration by re-establishing the incorrect configuration.

- Step: Replace original hardware components, Command: undo replace , Description: Restore the original hardware components by re-inserting the faulty components.